AI Quant Research Lab

Comprehensive Research Report

Research assets: AAPL and a selected universe of 10 US equities Research period: 2021–2025, with primary out-of-sample evaluation during 2024–2025 Report type: Historical quantitative research and strategy evaluation

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1. Executive Summary

This project investigates whether systematic trading strategies and machine-learning models can identify repeatable opportunities in historical equity-market data.

The research began with single-stock experiments on Apple Inc. (AAPL) and subsequently expanded to a selected ten-stock US equity universe.

The multi-stock universe includes technology, consumer, financial, energy and healthcare companies. The portfolio experiment uses equal-weighted daily rebalancing.

The research emphasizes out-of-sample evaluation, transaction costs, execution timing, statistical testing, parameter sensitivity and explicit rejection of strategies that fail to demonstrate convincing evidence.

The experiment registry currently contains 6 experiments, of which 6 are rejected.

The multi-stock mean-reversion experiment evaluated 10 stocks over the 2024–2025 out-of-sample period. None of the ten individual stocks outperformed their respective benchmarks on total return.

The equal-weighted portfolio also underperformed its benchmark. Its mean daily excess return was significantly negative under HAC inference, with a p-value of approximately 0.0086.

The findings do not establish a reliable benchmark-beating trading edge. Instead, the experiments demonstrate the importance of realistic execution assumptions, statistical validation and disciplined research methodology.

2. Research Objective

The objective is to develop a research framework that attempts to disprove its own trading hypotheses rather than simply identify historical strategies with attractive returns.

The framework evaluates:

• Signal generation and execution timing.

• Historical performance against buy-and-hold.

• Transaction-cost sensitivity.

• Out-of-sample performance.

• Statistical significance.

• Parameter sensitivity.

• Dependence on unusually strong trading days.

• Predictive performance of machine-learning models.

Research principle

A strategy is not accepted merely because it produces positive historical returns. It must demonstrate credible performance after costs, against an appropriate benchmark, and under out-of-sample evaluation.

3. Moving-Average Strategy

The moving-average strategy uses a short-term simple moving average (SMA) and a long-term SMA.

A long position is held when the short-term average is above the long-term average. The previous session's signal determines the current session's position.

The historical backtest uses adjusted closing prices and assumes a 0.10% transaction cost per position change.

Earlier parameter sensitivity experiment

An earlier experiment evaluated four parameter combinations: 10/30, 20/50, 30/100 and 50/150.

All configurations use a common evaluation start determined by the first valid 150-session moving average. The notebook's research period is January 2021 to December 2025.

Short MA	Long MA	Strategy Return	Strategy CAGR	Strategy Sharpe	Strategy Max Drawdown	Benchmark Return	Number of Position Changes
10	30	22.30%	4.69%	0.3408	-27.95%	89.30%	41

Short MA	Long MA	Strategy Return	Strategy CAGR	Strategy Sharpe	Strategy Max Drawdown	Benchmark Return	Number of Position Changes
20	50	29.88%	6.14%	0.4157	-29.80%	89.30%	26
30	100	33.24%	6.76%	0.4372	-31.41%	89.30%	12
50	150	11.79%	2.57%	0.2275	-28.40%	89.30%	11

The earlier parameter experiment is a historical sensitivity analysis, not an independent out-of-sample test.

Later six-configuration analysis

A separate analysis evaluated six moving-average configurations:

Short_Window	Long_Window	Total Return	CAGR	Annualized Volatility	Sharpe Ratio (rf=0)	Maximum Drawdown	Buy_Hold_Total_Return	Buy_Hold_Sharpe	Positive_Return	Beats_Buy_Hold_Return	Beats_Buy_Hold_Sharpe
10	30	28.77%	5.21%	17.94%	0.3723	-27.95%	115.81%	0.6932	True	False	False
15	40	38.51%	6.76%	17.28%	0.4648	-32.34%	115.81%	0.6932	True	False	False
20	50	33.15%	5.92%	17.83%	0.4113	-29.80%	115.81%	0.6932	True	False	False
25	60	67.38%	10.90%	17.77%	0.6709	-24.22%	115.81%	0.6932	True	False	False
30	100	44.69%	7.70%	18.23%	0.4979	-31.41%	115.81%	0.6932	True	False	False
50	150	11.79%	2.26%	18.52%	0.2136	-28.40%	115.81%	0.6932	True	False	False

The two parameter experiments should be interpreted separately. Their results should not be combined because their evaluation procedures and source data may differ.

Interpretation

The moving-average experiments did not establish consistent outperformance over buy-and-hold. Parameter sensitivity also demonstrates that historical performance depends on the selected moving-average windows.

The six-configuration analysis identified a stronger historical configuration, but it still did not outperform buy-and-hold on total return or Sharpe ratio.

4. Mean-Reversion Strategy — Close Execution

The close-execution mean-reversion experiment evaluates whether prices sufficiently below a rolling average subsequently recover.

Parameters were selected using the 2021–2023 training period and evaluated on 2024–2025 data.

The selected configuration was a 20-session window with a -5% entry threshold.

Out-of-sample performance

Metric	Mean Reversion	Buy and Hold
Total Return	8.72%	42.54%
CAGR	4.29%	19.47%
Annualized Volatility	15.18%	27.99%
Sharpe Ratio (rf=0)	0.3495	0.7742
Maximum Drawdown	-18.96%	-33.36%

HAC significance test

Observations	Mean_Daily_Excess_Return	Ordinary_T_Statistic	Ordinary_P_Value	HAC_Lags	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	HAC_CI_95_Lower	HAC_CI_95_Upper
502	-0.06%	-0.9808	0.3272	5	0.07%	-0.9266	0.3546	-0.20%	0.07%

Transaction-cost sensitivity

Transaction_Cost_Per_Turnover	Strategy_Total_Return	Strategy_CAGR	Strategy_Sharpe	Strategy_Max_Drawdown	Buy_Hold_Total_Return	Buy_Hold_Sharpe
0.00%	10.93%	5.34%	0.4162	-18.62%	42.54%	0.7742
0.05%	9.82%	4.82%	0.3828	-18.79%	42.54%	0.7742
0.10%	8.72%	4.29%	0.3495	-18.96%	42.54%	0.7742
0.25%	5.50%	2.72%	0.2496	-19.45%	42.54%	0.7742
0.50%	0.32%	0.16%	0.0841	-20.48%	42.54%	0.7742

The HAC p-value was approximately 0.3546. The experiment therefore did not demonstrate statistically significant mean daily excess returns at the 5% level.

The strategy also underperformed buy-and-hold on total return and Sharpe ratio.

Decision: REJECT.

5. Mean-Reversion Strategy — Open Execution

A separate implementation evaluates next-session execution using opening prices.

A signal observed at the previous close determines the position held during the next open-to-open return interval.

Parameters were selected using 2021–2023 training data and evaluated out of sample over 2024–2025.

The selected configuration was a 20-session window with a -5% entry threshold.

Out-of-sample performance

Metric	Mean Reversion (Open Execution)	Buy and Hold (Open-to-Open)
Total Return	11.50%	47.28%
CAGR	5.63%	21.50%
Annualized Volatility	17.21%	29.90%
Sharpe Ratio (rf=0)	0.4018	0.7998
Maximum Drawdown	-13.44%	-33.33%

The open-execution experiment produced a total return of approximately 11.50%, compared with 47.28% for its open-to-open buy-and-hold benchmark.

Its HAC p-value, as recorded in the experiment registry, was approximately 0.3404. This does not provide statistically significant evidence of positive mean daily excess returns at the 5% level.

Lower volatility and drawdown do not compensate for the substantial underperformance against the benchmark.

Decision: REJECT.

6. Static Machine-Learning Experiment

A logistic-regression classifier was evaluated using historical price-derived features.

The experiment investigated whether the model could predict the direction of subsequent returns.

Prediction metrics

Metric	Value
Accuracy	0.477
Balanced Accuracy	0.4623
Precision	0.5166
Recall	0.6264
ROC AUC	0.4662
Brier Score	0.2587
Training Majority Baseline Accuracy	0.5449
Always-Up Baseline Accuracy	0.5449
Historical-Rate Baseline Brier Score	0.2484

Strategy performance

Metric	ML Strategy	Buy and Hold
Total Return	3.24%	47.83%
CAGR	1.62%	21.73%
Annualized Volatility	23.44%	27.89%
Sharpe Ratio	0.1838	0.8429
Max Drawdown	-31.18%	-33.36%

The model's ROC AUC was approximately 0.4661, below the 0.50 level associated with random ranking.

The resulting strategy also underperformed buy-and-hold.

These results do not support the use of this model as a profitable trading signal in its tested configuration.

7. Walk-Forward Validation

Walk-forward validation evaluates a model sequentially using historical information available before each prediction.

This is a more realistic evaluation design than fitting a model once and evaluating it on a fixed holdout period.

Prediction metrics

Metric	Value
Accuracy	0.513
Balanced Accuracy	0.4848
Precision	0.5356
Recall	0.7985
ROC AUC	0.4469
Brier Score	0.2542
Expanding Majority Baseline Accuracy	0.5449
Historical-Rate Baseline Brier Score	0.2485

Strategy performance

Metric	Walk-Forward ML	Buy and Hold
Total Return	13.00%	47.83%
CAGR	6.34%	21.73%
Annualized Volatility	25.38%	27.89%
Sharpe Ratio	0.3676	0.8429
Max Drawdown	-31.73%	-33.36%

The walk-forward model did not establish predictive performance above its simple baselines. Its trading strategy also underperformed buy-and-hold.

Walk-forward evaluation improves the testing design, but does not by itself establish that a model is free from data leakage or implementation bias. Those properties depend on the complete feature-generation and execution pipeline.

8. Strategy Diagnostics and Multi-Stock Research

Earlier moving-average best-day dependency

The earlier SMA experiment replaced its five strongest daily strategy returns with zero and recalculated performance.

This is an observation-deletion sensitivity test, not an executable alternative strategy.

Metric	Original Strategy	Without Best 5 Days
Total Return	26.54%	-0.68%
CAGR	12.54%	-0.34%
Annualized Volatility	18.52%	16.67%
Sharpe Ratio (0% risk-free)	0.7304	0.0631
Maximum Drawdown	-25.03%	-25.03%

Mean-reversion best-day dependency

A separate diagnostic examined the effect of removing the strongest daily returns from the close-execution mean-reversion strategy and its benchmark.

Series	Test	Days_Removed	Total_Return	Return_Change	Observations_Remaining
Mean Reversion Strategy	Baseline	0	8.72%	0.00%	502
Mean Reversion Strategy	Remove top 1 days	1	-5.73%	-14.45%	501
Mean Reversion Strategy	Remove top 3 days	3	-12.30%	-21.03%	499
Mean Reversion Strategy	Remove top 5 days	5	-17.02%	-25.75%	497
Mean Reversion Strategy	Remove top 10 days	10	-25.10%	-33.83%	492
Buy and Hold	Baseline	0	42.54%	0.00%	502
Buy and Hold	Remove top 1 days	1	23.59%	-18.95%	501
Buy and Hold	Remove top 3 days	3	8.38%	-34.16%	499
Buy and Hold	Remove top 5 days	5	-2.69%	-45.23%	497
Buy and Hold	Remove top 10 days	10	-20.69%	-63.23%	492

The two dependency analyses use different procedures and should not be directly compared.

Extreme-day analysis

The following table summarizes five of the strongest and five of the weakest recorded daily observations.

Extreme_Group	Date	Position	Asset_Return_Pct	Gross_Return_Pct	Transaction_Cost_Pct	Net_Return_Pct
Best 10	2025-04-09	1	15.33%	15.33%	0.00%	15.33%
Best 10	2025-04-11	1	4.06%	4.06%	0.00%	4.06%
Best 10	2025-04-22	1	3.41%	3.41%	0.10%	3.31%
Best 10	2025-01-27	1	3.18%	3.18%	0.00%	3.18%
Best 10	2025-05-27	1	2.53%	2.53%	0.10%	2.43%
Worst 10	2025-03-13	1	-3.36%	-3.36%	0.00%	-3.36%
Worst 10	2025-04-07	1	-3.67%	-3.67%	0.00%	-3.67%
Worst 10	2025-04-10	1	-4.24%	-4.24%	0.00%	-4.24%
Worst 10	2025-04-08	1	-4.98%	-4.98%	0.00%	-4.98%
Worst 10	2025-04-04	1	-7.29%	-7.29%	0.10%	-7.39%

April 2025 market-regime observations

The following table selects the eight April 2025 sessions with the largest absolute underlying asset moves. It includes the strategy's signal, position, transaction cost and net return where those fields are available.

These are descriptive observations. They do not establish causal explanations for strategy performance.

Date	Asset_Return_Pct	Signal	Position	Transaction_Cost_Pct	Net_Return_Pct	Strategy_Equity	Buy_Hold_Equity
2025-04-03	-9.25%	1	0	0.00%	-0.00%	1.0229	1.0791
2025-04-04	-7.29%	1	1	0.10%	-7.39%	0.9473	1.0004
2025-04-07	-3.67%	1	1	0.00%	-3.67%	0.9125	0.9637
2025-04-08	-4.98%	1	1	0.00%	-4.98%	0.867	0.9157
2025-04-09	15.33%	1	1	0.00%	15.33%	1	1.056
2025-04-10	-4.24%	1	1	0.00%	-4.24%	0.9576	1.0113
2025-04-11	4.06%	1	1	0.00%	4.06%	0.9964	1.0523
2025-04-16	-3.89%	1	0	0.00%	-0.00%	1.0174	1.0317

Multi-Stock Mean-Reversion Research

The research universe was expanded from AAPL to ten selected US equities:

AAPL, MSFT, NVDA, GOOGL, AMZN, WMT, JPM, GS, XOM and JNJ.

All ten datasets passed market-data validation and had aligned trading dates. The common dataset contained 1,261 observations per stock, covering 2021-01-04 through 2026-01-09.

The mean-reversion strategy used the 20-session rolling window and -5% entry threshold selected using AAPL training data from 2021–2023. These parameters were held fixed across the ten-stock evaluation. The out-of-sample period was 2024–2025.

The use of AAPL-selected parameters across other stocks is a deliberate fixed-parameter transfer experiment. It is not independent parameter optimization for each stock.

Individual-stock performance

ticker	company	sector	window	threshold	transaction_cost	test_start	test_end	observations	Strategy_Total Return	Strategy_CAGR	Strategy_Annualized Volatility	Strategy_Sharpe Ratio (rf=0)	Strategy_Maximum Drawdown	BuyHold_Total Return	BuyHold_CAGR	BuyHold_Annualized Volatility	BuyHold_Sharpe Ratio (rf=0)	BuyHold_Maximum Drawdown	Excess_Total_Return	Beats_Buy_Hold
AAPL	Apple	Technology	20	-0.05	0.10%	2024-01-02	2025-12-30	501	11.50%	5.63%	17.21%	0.4018	-13.44%	47.28%	21.50%	29.90%	0.7998	-33.33%	-35.78%	False
MSFT	Microsoft	Technology	20	-0.05	0.10%	2024-01-02	2025-12-30	501	8.39%	4.13%	10.12%	0.45	-8.46%	32.43%	15.18%	24.31%	0.7019	-24.43%	-24.05%	False
NVDA	NVIDIA	Technology	20	-0.05	0.10%	2024-01-02	2025-12-30	501	119.12%	48.38%	28.40%	1.528	-12.94%	285.18%	97.05%	53.50%	1.5378	-42.84%	-166.06%	False
GOOGL	Alphabet	Technology	20	-0.05	0.10%	2024-01-02	2025-12-30	501	12.98%	6.33%	11.58%	0.587	-9.43%	127.51%	51.21%	32.85%	1.4213	-30.32%	-114.53%	False
AMZN	Amazon	Consumer	20	-0.05	0.10%	2024-01-02	2025-12-30	501	20.31%	9.75%	17.89%	0.6066	-14.48%	53.70%	24.13%	33.83%	0.8075	-32.22%	-33.39%	False
WMT	Walmart	Consumer	20	-0.05	0.10%	2024-01-02	2025-12-30	501	-0.92%	-0.46%	4.78%	-0.073	-5.92%	117.78%	47.92%	21.86%	1.9004	-23.59%	-118.69%	False
JPM	JPMorgan Chase	Financials	20	-0.05	0.10%	2024-01-02	2025-12-30	501	10.03%	4.92%	10.47%	0.5098	-9.71%	100.55%	41.91%	24.09%	1.5741	-25.71%	-90.53%	False
GS	Goldman Sachs	Financials	20	-0.05	0.10%	2024-01-02	2025-12-30	501	8.80%	4.33%	14.81%	0.3592	-13.16%	141.25%	55.73%	29.30%	1.659	-32.88%	-132.45%	False
XOM	Exxon Mobil	Energy	20	-0.05	0.10%	2024-01-02	2025-12-30	501	2.53%	1.26%	8.01%	0.1965	-5.77%	28.73%	13.54%	23.06%	0.6669	-19.30%	-26.20%	False

ticker	company	sector	window	threshold	transaction_cost	test_start	test_end	observations	Strategy_Total Return	Strategy_CAGR	Strategy_Annualized Volatility	Strategy_Sharpe Ratio (rf=0)	Strategy_Maximum Drawdown	BuyHold_Total Return	BuyHold_CAGR	BuyHold_Annualized Volatility	BuyHold_Sharpe Ratio (rf=0)	BuyHold_Maximum Drawdown	Excess_Total_Return	Beats_Buy_Hold
JNJ	Johnson and Johnson	Healthcare	20	-0.05	0.10%	2024-01-02	2025-12-30	501	4.45%	2.21%	7.22%	0.3399	-7.42%	40.19%	18.52%	17.88%	1.0401	-15.40%	-35.74%	False

All ten stocks underperformed their respective benchmarks on total return at the reference transaction cost of 0.10%.

Cross-sectional results

Metric	Mean	Median	Standard Deviation	Minimum	25th Percentile	75th Percentile	Maximum
Strategy Total Return	19.72%	9.41%	35.42%	-0.92%	5.43%	12.61%	119.12%
Buy-and-Hold Total Return	97.46%	77.12%	78.28%	28.73%	41.96%	125.08%	285.18%
Excess Total Return	-77.74%	-63.15%	52.72%	-166.06%	-117.65%	-33.97%	-24.05%
Strategy Sharpe Ratio	0.4906	0.4259	0.4154	-0.0730	0.3447	0.5677	1.5280
Buy-and-Hold Sharpe Ratio	1.2109	1.2307	0.4565	0.6669	0.8017	1.5650	1.9004
Strategy Maximum Drawdown	-10.07%	-9.57%	3.24%	-14.48%	-13.11%	-7.68%	-5.77%
Buy-and-Hold Maximum Drawdown	-28.00%	-28.02%	7.94%	-42.84%	-32.71%	-23.80%	-15.40%

Sector-level descriptive results

sector	Stocks	Mean_Strategy_Return	Median_Strategy_Return	Mean_BuyHold_Return	Median_BuyHold_Return	Mean_Excess_Return	Median_Excess_Return	Mean_Strategy_Sharpe	Mean_BuyHold_Sharpe	Stocks_Beating_BuyHold	Percent_Beating_BuyHold
Consumer	2	9.70%	9.70%	85.74%	85.74%	-76.04%	-76.04%	0.2668	1.3539	0	0
Energy	1	2.53%	2.53%	28.73%	28.73%	-26.20%	-26.20%	0.1965	0.6669	0	0
Financials	2	9.41%	9.41%	120.90%	120.90%	-111.49%	-111.49%	0.4345	1.6165	0	0
Healthcare	1	4.45%	4.45%	40.19%	40.19%	-35.74%	-35.74%	0.3399	1.0401	0	0
Technology	4	38.00%	12.24%	123.10%	87.40%	-85.10%	-75.16%	0.7417	1.1152	0	0

Sector groups are small and unevenly represented. These descriptive results should not be interpreted as reliable sector-wide conclusions.

Transaction-cost sensitivity

The experiment was reconstructed at several transaction-cost assumptions. The 0.10% reconstruction was checked against the original strategy results.

transaction_cost	stocks	Mean_Strategy_Return	Median_Strategy_Return	Mean_Excess_Return	Median_Excess_Return	Mean_Strategy_Sharpe	Median_Strategy_Sharpe	Mean_Strategy_Max_Drawdown	Stocks_Beating_BuyHold	Percent_Beating_BuyHold
0.00%	10	22.02%	10.95%	-75.44%	-62.06%	0.5557	0.4898	-9.78%	0	0
0.05%	10	20.86%	10.18%	-76.60%	-62.59%	0.5232	0.4579	-9.93%	0	0
0.10%	10	19.72%	9.41%	-77.74%	-63.15%	0.4906	0.4259	-10.07%	0	0
0.25%	10	16.37%	7.15%	-81.08%	-65.79%	0.3926	0.3297	-10.64%	0	0
0.50%	10	11.06%	2.27%	-86.40%	-70.04%	0.2297	0.1539	-11.74%	0	0

Even with zero transaction costs, none of the ten stocks outperformed its benchmark. Transaction costs worsened the results but did not explain the underlying underperformance.

Individual-stock HAC significance tests

The individual-stock tests evaluate whether each stock's mean daily strategy excess return differs from zero.

Holm correction is applied across the ten individual tests to control the family-wise error rate.

ticker	Observations	Mean_Daily_Excess_Return	Annualized_Mean_Excess_Return	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	CI_95_Lower	CI_95_Upper	Holm_Adjusted_P_Value	Significant_After_Holm	Raw_Conclusion	Holm_Conclusion
WMT	501	-0.17%	-41.89%	0.06%	-2.8778	0.004	-0.28%	-0.05%	0.0401	True	Significant	Significant after Holm correction
GS	501	-0.17%	-43.28%	0.07%	-2.349	0.0188	-0.32%	-0.03%	0.1694	False	Significant	Not significant after Holm correction
JPM	501	-0.13%	-32.58%	0.06%	-2.0373	0.0416	-0.25%	-0.00%	0.3329	False	Significant	Not significant after Holm correction
GOOGL	501	-0.16%	-39.89%	0.08%	-1.9877	0.0469	-0.31%	-0.00%	0.3329	False	Significant	Not significant after Holm correction
JNJ	501	-0.06%	-16.14%	0.05%	-1.3655	0.1721	-0.16%	0.03%	1	False	Not significant	Not significant after Holm correction

ticker	Observations	Mean_Daily_Excess_Return	Annualized_Mean_Excess_Return	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	CI_95_Lower	CI_95_Upper	Holm_Adjusted_P_Value	Significant_After_Holm	Raw_Conclusion	Holm_Conclusion
NVDA	501	-0.15%	-38.88%	0.13%	-1.183	0.2368	-0.41%	0.10%	1	False	Not significant	Not significant after Holm correction
AAPL	501	-0.07%	-17.00%	0.07%	-0.9542	0.34	-0.21%	0.07%	1	False	Not significant	Not significant after Holm correction
XOM	501	-0.05%	-13.80%	0.06%	-0.8953	0.3706	-0.17%	0.07%	1	False	Not significant	Not significant after Holm correction
MSFT	501	-0.05%	-12.51%	0.06%	-0.8554	0.3924	-0.16%	0.06%	1	False	Not significant	Not significant after Holm correction
AMZN	501	-0.07%	-16.47%	0.08%	-0.7801	0.4353	-0.23%	0.10%	1	False	Not significant	Not significant after Holm correction

Four stocks had unadjusted p-values below 0.05. After Holm correction, only WMT remained significant at the 5% level. Its significant result represents negative mean daily excess return, not evidence of a profitable strategy.

Equal-weighted portfolio HAC test

The portfolio is constructed by averaging the ten constituent strategy returns each day. The benchmark is constructed by averaging the ten constituent asset returns each day. Both are therefore equal-weighted and daily rebalanced.

Observations	Mean_Daily_Excess_Return	Annualized_Mean_Excess_Return	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	CI_95_Lower	CI_95_Upper	Stocks	Portfolio_Construction	Conclusion
501	-0.11%	-27.24%	0.04%	-2.6259	0.0086	-0.19%	-0.03%	10	Equal-weighted daily rebalanced	Significant

The portfolio contained 501 daily observations. Its mean daily excess return was approximately -0.1081%, with a HAC p-value of 0.008642. The 95% confidence interval was entirely negative.

HAC lag sensitivity

HAC_Max_Lags	Observations	Mean_Daily_Excess_Return	Annualized_Mean_Excess_Return	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	CI_95_Lower	CI_95_Upper	Significant_At_5_Percent
1	501	-0.11%	-27.24%	0.04%	-2.6981	0.007	-0.19%	-0.03%	True
5	501	-0.11%	-27.24%	0.04%	-2.6259	0.0086	-0.19%	-0.03%	True
10	501	-0.11%	-27.24%	0.04%	-2.541	0.0111	-0.19%	-0.02%	True
20	501	-0.11%	-27.24%	0.04%	-2.5138	0.0119	-0.19%	-0.02%	True

The negative portfolio excess-return result remained statistically significant at the 5% level across HAC lag specifications of 1, 5, 10 and 20.

Equal-weighted portfolio performance

The equal-weighted portfolio's compounded results were:

Metric	Strategy	Equal-weighted benchmark
Total return	17.69%	97.34%
Sharpe ratio	0.8649	1.9342
Maximum drawdown	-5.39%	-22.45%

The strategy experienced a smaller maximum drawdown, but substantially underperformed the benchmark on total return and Sharpe ratio.

Decision: REJECT.

Interpretation

The multi-stock experiment provides broader evidence than the original AAPL-only research, but it does not establish a profitable or benchmark-beating mean-reversion strategy.

The ten-stock universe is selected rather than a point-in-time historical universe. The findings are conditional on the selected assets, fixed AAPL-derived parameters, transaction-cost assumptions and 2024–2025 evaluation period.

9. Experiment Registry

The registry records formal strategy experiments and their decisions.

Experiment_ID	Strategy	Evaluation_Period	Strategy_Total_Return	Buy_Hold_Total_Return	Strategy_Sharpe	Buy_Hold_Sharpe	HAC_P_Value	Decision
AAPL_MR_OOS_2024_2025	Mean Reversion	2024-2025	8.72%	42.54%	0.3495	0.7742	0.3546	REJECT
AAPL_SMA_BASELINE	Moving Average	2021-2025	33.15%	115.81%	0.4113	0.6932	Not available	REJECT
AAPL_MR_OPEN_OOS_2024_2025	Mean Reversion (Open Execution)	2024-2025	11.50%	47.28%	0.4018	0.7998	0.3404	REJECT
US10_MR_OPEN_PORTFOLIO_OOS_2024_2025	Mean Reversion (Open Execution)	2024-2025	17.69%	97.34%	0.8649	1.9342	0.0086	REJECT
AAPL_STATIC_ML_OOS_2024_2025	Static Logistic Regression	2024-2025	3.24%	47.83%	0.1838	0.8429	Not available	REJECT
AAPL_WALK_FORWARD_ML_OOS_2024_2025	Walk-Forward Logistic Regression	2024-2025	13.00%	47.83%	0.3676	0.8429	Not available	REJECT

The registry currently contains 6 experiments, of which 6 are rejected.

The registry includes five AAPL experiments and one ten-stock equal-weighted portfolio experiment.

The moving-average registry entry does not have a verified HAC significance result. Its unavailable p-value is not interpreted as evidence of statistical significance.

The close-execution and open-execution AAPL mean-reversion experiments did not demonstrate statistically significant mean daily excess returns at the 5% level.

The ten-stock portfolio experiment recorded a HAC p-value of approximately 0.0086, indicating significantly negative mean daily excess returns at the 5% level. This result remained significant across the tested HAC lag specifications.

The portfolio's statistical result is conditional on the selected ten-stock universe and 2024–2025 evaluation period.

10. Limitations

The research has several important limitations:

1. The initial experiments use AAPL. The later multi-stock experiment expands the universe to ten selected US equities, but this is not a broad-market sample.

2. The ten-stock universe is selected using a present-day configuration rather than a point-in-time historical constituent universe. Survivorship and selection bias may therefore affect the findings.

3. The universe is not sector-balanced. Technology accounts for four of the ten stocks, while several other sectors have only one or two representatives. Sector-level statistics are descriptive, not generalizable.

4. Multi-stock mean-reversion parameters were selected using AAPL training data and transferred unchanged to the other stocks. This does not establish that the parameters are optimal for each constituent.

5. The portfolio benchmark is an equal-weighted, daily rebalanced basket. It should not be interpreted as a conventional buy-and-hold portfolio with fixed share quantities.

6. Transaction-cost assumptions simplify real-world execution. Market impact, bid-ask spreads, liquidity constraints and partial fills are not fully modeled.

7. Some experiments use different execution conventions and evaluation periods, so their returns are not directly comparable.

8. Parameter sensitivity and repeated experimentation do not eliminate data-snooping risk or multiple-testing concerns.

9. Machine-learning features are primarily derived from historical price data. Further independent validation is required.

10. The HAC tests evaluate mean daily arithmetic excess returns. They do not directly test whether compounded total returns or CAGR are statistically different.

11. The annualized mean excess return reported in the HAC analysis is the arithmetic daily mean multiplied by 252. It is not a compounded annual return.

12. The framework is a research prototype, not a production trading system.

11. Future Research

Potential extensions include:

• Expanding to a broader, point-in-time US equity universe.

• Controlling survivorship and selection bias.

• Testing across multiple assets and market regimes.

• Evaluating parameters independently using nested or properly separated training and validation procedures.

• Adding point-in-time fundamental and alternative data.

• Modeling spreads, slippage and market impact.

• Applying purged and embargoed cross-validation where appropriate.

• Evaluating multiple-testing and data-snooping adjustments.

• Testing portfolio-level risk and exposure constraints.

• Improving feature engineering and model calibration.

• Comparing against stronger statistical and economic baselines.

• Requiring independent validation before accepting a strategy.

12. Final Conclusion

The experiments did not identify a strategy that convincingly outperformed its benchmark after the tested transaction costs and evaluation procedures.

The single-stock and multi-stock mean-reversion experiments underperformed their respective benchmarks. The equal-weighted ten-stock portfolio also produced significantly negative mean daily excess returns under the tested HAC specifications.

The machine-learning experiments failed to demonstrate reliable predictive superiority over simple baselines.

The main contribution of this project is its research framework: strategies are tested under explicit assumptions, evaluated against benchmarks, examined for statistical support and rejected when the evidence is insufficient.

The results should be interpreted as historical research findings, not investment advice or evidence of a deployable trading edge.